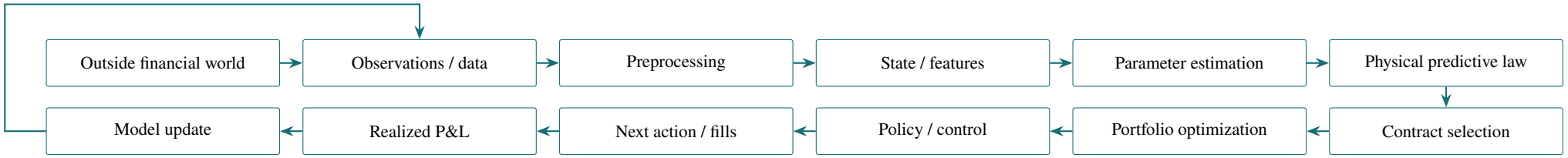
SPY options: thesis paths under P, option values under Q

Baseline: one-period Heston/utility MPC. Extension: a slow valuation anchor, a fixed economic deadline and adaptive expression.



Keep the physical thesis separate from its current instrument. Observations update beliefs; strategic decisions and child-order execution use separate clocks.

Three clocks. Observations at o_j may be intraday; strategic decisions at t_k may be daily; the economic horizon ends at a fixed T^{th}, e.g. four weeks. H is the number of planned decision intervals, with $t_{k+H} \le T^{\text{th}}$. Released information is $\mathcal{F}_k$. Freeze priors, proxy definitions, calendar/surprise laws, fits, bounds, simulation count P, seed, numerical grids and test blocks. Between decisions observe/filter; only scheduled decisions or mandatory risk/exercise events authorize portfolio changes.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O0. Released data / features $S, Y^v > 0$; J listed legs with strike K_j, expiry T_j, multiplier m_j, sign $\chi_j = \pm 1$. $C, \omega \in \mathbb{R}^J$, $X \in \mathbb{R}^{2+2J}$; economic proxy $y \in \mathbb{R}$.	Synchronize spot, bid/ask, dividends, funding, liquidity and notices; reject stale/crossed quotes. For an interval of Δ years set $Y^v = \max\{\epsilon_v, \sum_l (\Delta_l \log S)^2/\Delta\}$. Dollar midpoint $C_j = m_j(\text{bid}_j + \text{ask}_j)/2$; $\omega_j = \max\{\omega_{\min}, m_j(\text{ask}_j - \text{bid}_j)/2\}$, positive floors. $X = (S, \log Y^v, C, \omega)$, plus mask/calendar. Map released economic evidence to a predeclared log-valuation proxy y; record its formula and timestamps. **To O1–O4.**	Records; NumPy `(J,)`. Optional option grid `(N_K,N_T,2)` plus mask. Economic evidence is data, not a retrospectively revised story.
O1. Fast market / slow thesis $Z = (S, v, u) \in \mathbb{R}^2_+ \times \mathbb{R}$; u: latent log-valuation anchor; $d = u - \log S$: convergence gap. $c \in \mathbb{R}^P$: catalyst surprises. $\theta \in \Theta \subset \mathbb{R}^{12+P}$: seven Heston parameters plus slow coefficients.	Between decision dates, freeze u_k and use $\mathrm{d}S/S = \mu_k^P(S, u_k)\,\mathrm{d}t + \sqrt{v}\,\mathrm{d}W_1^P$, $\mathrm{d}v = \kappa_P(\bar{v}_P - v)\,\mathrm{d}t + \xi\sqrt{v}(\rho\,\mathrm{d}W_1^P + \sqrt{1-\rho^2}\,\mathrm{d}W_2^P)$, $\mu_k^P = \mu + \kappa_d(u_k - \log S)$. Slow model: $u_{k+1} = \bar{u} + \phi(u_k - \bar{u}) + \beta^\top c_{k+1} + \sigma_u \eta$, $y = u + \tau_y \nu$, $\beta \in \mathbb{R}^P$. Independent $\eta, \nu \sim \mathcal{N}(0,1)$; fit $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$. At announced releases $c \sim \mathcal{N}(\bar{c}, C_c)$, fitted to past surprises; otherwise $c = 0$. Under $\mathbb{Q}$, stock drift is r, variance drift $\kappa_Q(\bar{v}_Q - v)$; ξ, ρ are shared. Risk prices: $\lambda_1 = (\mu_k^P - r)/\sqrt{v}$, $\xi\sqrt{v}(\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2) = \kappa_P(\bar{v}_P - v) - \kappa_Q(\bar{v}_Q - v)$. Impose $2\kappa_P\bar{v}_P, 2\kappa_Q\bar{v}_Q \ge \xi^2$, $\lvert\rho\rvert < 1$, positive variance coefficients and admissible moment/martingale bounds. Apply announced cash-dividend jumps on the stated positive-price domain. $\kappa_d = 0$ recovers constant drift. **To O2–O3.**	Fast `(2,)` / slow scalar; parameter vector `(12+p,)`. Conditional transitions are callables. Only $\mathbb{P}$ forecasts thesis convergence; the anchor is not a $\mathbb{Q}$ stock price.
O2. Filter / estimate $b_k \in \mathcal{P}(\mathbb{R}_+ \times \mathbb{R} \times \Theta)$: posterior of (v, u, θ). $\widehat{\theta}, \widehat{v}, \widehat{u}$: mode; ℓ_k: measurement likelihood.	Start with a fixed proper prior. Measurements: $\log Y^v \mid v \sim \mathcal{N}(\log v, \sigma_v^2)$, $y \mid u \sim \mathcal{N}(u, \tau_y^2)$, $C_j \mid Z, \theta \sim \mathcal{N}(c_j^A, \omega_j^2)$, conditionally independent; omit missing observations. With the O1 joint transition, $b_{k+1}(v, u, \theta) \propto \ell_{k+1}(v, u, \theta) \iint p_\theta^P(S_{k+1}, v, u \mid S_k, v', u') b_k(v', u', \theta)\,\mathrm{d}v'\,\mathrm{d}u'$. Normalize; use quadrature/particles and update released measurements between decisions. Parameters remain fixed along each future path; decisions integrate the posterior, not just its mode. Mode Greeks are a local constraint approximation. **To O3–O5.**	Weighted particles `(N_particles,14+p)` or quadrature; fit by likelihood/Bayes update. Observation noise $\sigma_v > 0$; retain posterior uncertainty.
O3. Path forecast then reprice P_k^H: physical law on $Z_{k:k+H}$. $c_j^A \in \mathbb{R}_+$: dollars/contract; $G \in \mathbb{R}^{3\times J}$: delta/gamma/vega. $g_j(S) = [\chi_j(S - K_j)]_+$.	$\mathsf{P}_k^H = \int \text{Law}_\theta^P(Z_{k:k+H} \mid S_k, v, u) b_k(\mathrm{d}v, \mathrm{d}u, \mathrm{d}\theta)$. $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only its endpoint marginal. Draw real-world *paths*, catalyst outcomes and future noisy observations. At *each* simulated state $Z_l^{(p)}$, mark each live option with the paired $\mathbb{Q}$ Snell solver: $c_j^A(l, S, v) = \max\{m_j g_j(S), e^{-\int_l^{l'} r\,\mathrm{d}t} \mathbb{E}_\theta^Q[c_j^A(l', S', v') \mid S, v]\}$, terminal $m_j g_j$. Refine the exercise grid (including dividends/expiry) to a fixed tolerance. $G_{\cdot j} = (\partial_S c_j^A, \partial_{SS} c_j^A, 2\sqrt{v}\partial_v c_j^A)^\top$. Thus $\mathbb{P}$ weights future outcomes; $\mathbb{Q}$ gives conditional option marks, never probabilities of thesis success. Re-filter synthetic observations before each future action. **To O5–O7.**	Paths `(P,H+1,3)` (subdivide for diffusion/events); marks `(P,H+1,J)`, Greeks `(3,J)`; price/filter callables.
O4. Thesis mode / instruments $p_k = b_k(d_k > \delta)$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed; $p_{\text{on}} > p_{\text{off}}$. $E_k \in \{0,1\}^{m\times J}$: dated leg selector; chosen $C \in \mathbb{R}^m$, $G \in \mathbb{R}^{3\times m}$.	Activate at $p \ge p_{\text{on}}$; retain between thresholds. Exit on convergence $\mathbb{E}d \le \delta_{\text{exit}}$, $p \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or fixed T^{th}; freeze thresholds and distinguish reasons. Waiting forbids new thesis risk. Closed † cancels entries and flattens thesis exposure, retaining delivery hedges; it is absorbing until a new record. Select live legs by quotes, tenor, liquidity/spread caps and delivery terms; retain held legs for exit. Use $EC, GE^\top$. Recompare *shares, calls, spreads, hedges and zero*. Expiry must cover holding or have a roll/expiry branch. An active thesis may hold zero. Baseline disables this gate and sets $\kappa_d = 0$. **To O5–O6.**	Probability/mode/deadline scalars; indices `(m,)`, selected arrays. Contract expiry, planning horizon and thesis deadline are distinct.

SPY options: dynamic policy and a multi-week example

The action is an instrument/hedge/exercise decision conditioned on a thesis belief. Assignment and future price paths remain random.

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O5. Baseline and longer control Account $s^{\text{acct}} = (q, h, B, \mathfrak{p})$, $q \in \mathbb{Z}^m$, $h \in l_h\mathbb{Z}$, $B \in \mathbb{R}$; $\mathfrak{p}$: pending claims. $a = (\Delta q, \Delta h, e^+)$: bounded integer/lot trades and owned exercise. $s = (b, z, T^{\text{th}} - t, s^{\text{acct}}, \text{pending orders})$.	Baseline $H = 1$: $a^* = \arg\max_{a \in \mathcal{A}(s)} P^{-1} \sum_p U(W_{k+1}^{\text{liq},(p)}(a))$, $U(w) = w - \lambda_U \sqrt{a_U^2 + w^2}$, $a_U > 0, 0 < \lambda_U < 1$. The ledger below generates each wealth sample. For $H > 1$, $Q_k^h(s,a) = \mathbb{E}_k^P[-\eta_T \|D_T a_{\text{trade}}\|_1 + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}(s)}[Q_k^h(s,a) - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}]$. At the deadline set $V^0 = U(W^{\text{liq}})$; earlier boundaries use this liquidation approximation or an estimated continuation value. One-period baseline has $\eta_T = h_0 = 0$. Expectations also average assignment/fills. Stochastic MPC executes the first action of a nonanticipative tree; full Bellman control plans to the deadline. Future actions see observations/posteriors, not latent states or later shocks. **To O6–O7.**	Finite action enumeration or global optimizer; CVXPY arrays for convex subproblems. D_T is a fixed positive diagonal trade-unit scale; $\eta_T, h_0 \geq 0$ are noncash preferences.
O5a. Risk / no-action region $\mathcal{A}(s)$: feasible controls; $b_G \in \mathbb{R}_+^3$, $\bar{q} \in \mathbb{Z}_+^m$. $\Pi_p = W_p^{\prime\text{liq}} - R_f W$, $R_f = e^{\int r\,dt}$; $\alpha \in (0,1)$.	At every node/interim event enforce $\lvert q\rvert \leq \bar{q}$, $0 \leq e^+ \leq (q + \Delta q)_+$, legal exercise dates, $\lvert Gq + (h + h^{\text{pend}}, 0, 0)^\top\rvert \leq b_G$, and available cash minus broker margin $\geq R_{\text{cash}}$, including joint-assignment stresses. One-step tail constraint: $\widehat{\text{ES}}_\alpha(-\Pi) = \min_\zeta\{\zeta + [P(1-\alpha)]^{-1} \sum_p [-\Pi_p - \zeta]_+\} \leq L_{\text{ES}}$. If hold is feasible, the full no-action region is $\max_{a \neq 0}[Q^H(s,a) - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}] \leq Q^H(s,0)$, also comparing owned exercise. Cash costs are already in wealth; turnover/switch penalties add inertia. Mandatory exercise/risk exits and O4 closure override waiting. Use fixed ties. **To O5 feasible actions.**	P&L samples `(P,)` per action; conditional mean/variance by sample moments; ES hinges; record value of holding and best alternative.
O6. Optimal execution is separate $y_e, v_e \in \mathbb{R}^{m+1}$: remaining parent trade and child rate; $T_e \ll \Delta_d$. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{(m+1)\times(m+1)}$.	Given O5's trade target, minimize $\int_0^{T_e} [s_e^\top \lvert v_e\rvert + v_e^\top \Lambda_e v_e + \lambda_e y_e^\top \Omega_e y_e]\,du$, $\dot{y}_e = -v_e$, $y_e(0) = a_{\text{trade}}$, $y_e(T_e) = 0$, with lot/rate/legging/margin limits. Fit impact from fills and Ω_e from short-horizon quote moves. Replace O5's forecast cash costs by expected execution costs; its risk penalty is not cash. Immediate full fills remain baseline. Pending/unfilled legs return to strategic state; owned-exercise instructions bypass this trade scheduler. **To O7 fills/exercise.**	Execution optimizer/callable, arrays `(m+1,)`, order book and fill records. Allocation selects exposure; execution schedules it.
O7. Cash, delivery and assignment $K \geq 0$: actual trade cost; $e = e^+ - A \in \mathbb{Z}^m$. A_j: assigned shorts; $\beta_A, f_j \in \mathbb{R}^5$. $B^{\text{pend}}, h^{\text{pend}} \in \mathbb{R}$: unsettled cash/shares.	For actual fills: $B' = B - C^\top \Delta q - S\Delta h - K$, $q' = q + \Delta q, h' = h + \Delta h$. On exercise, $q'' = q' - e$, $\Delta h^{\text{pend}} = \sum m_j \chi_j e_j$, $\Delta B^{\text{pend}} = -\sum m_j K_j \chi_j e_j$. Transfer at delivery once. Pre-expiry baseline $A_j \sim \text{Binom}((-q_j')_+, \sigma(\beta_A^\top f_j))$, independent conditionally; $\sigma(x) = (1 + e^{-x})^{-1}$, $f = (1, \log(S/K), T - t, C - mg, D_{\text{next}}^{\text{div}})$. Fit bounded β_A by binomial likelihood. At expiry use the declared exercise/contrary-notice rule; remove unexercised legs. Notices are immediate; delivery follows its calendar. Cash accrues as $B \leftarrow R_\ell[B]_+ - R_b[-B]_+ - \text{Borrow} + h^{\text{ent}} D^{\text{div}}$. $W = B + B^{\text{pend}} + (h + h^{\text{pend}})S + q^\top C$. Liquidate at executable costs for W^{liq}; actual future marks replace predicted $\mathbb{Q}$ marks. Baseline $K = s^\top \lvert a_{\text{trade}}\rvert + a_{\text{trade}}^\top \Lambda a_{\text{trade}}$, $\Lambda \succeq 0$: observed spreads/fees and fitted impact, held fixed in forecasts. **To O5 and next posterior/account.**	Event ledger / pending-claim records, assignment samples `(P,H,m)`; cash/borrow/entitlement schedules as supplied functions. Only owned exercise is controlled.
O8. RL and validation $Q_\psi : \mathbb{R}^{d_s + d_a} \to \mathbb{R}$, $d_a = 2m + 1$; $\varphi_s(s) \in \mathbb{R}^{d_s}$: sufficient finite belief/account encoding.	Optional RL replaces O5's value solve: target $Y = -\eta_T \|D_T a_{\text{trade}}\|_1 - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')$, terminal maximum replaced by $U(W^{\text{liq}})$; loss $\mathbb{E}[(Q_\psi(s,a) - Y)^2]$. Useful for expensive repeated solves or learned action-dependent fills/information; unnecessary for tractable identified MPC. Require supported actions and validated simulator/offline data; preserve hard constraints. Seal chronological tests of path/catalyst calibration, $\mathbb{P}/\mathbb{Q}$ errors, assignment, costs, turnover and net benchmark gains. Blocks must preserve horizon overlap and longer dependence; adjust multiplicity. Gate solves only for valid/nonempty problems; otherwise funded exit or stop/default. **To control/update gate.**	Model callable / tensors; training tuples (s, a, r, s'). Audit posterior, gap, thesis mode, held-vs-switch value, instrument and exit reason. RL does not create the thesis forecast.

Worked conceptual example (hypothetical, normalized prices). Daily decisions; intraday observations; deadline day 20; a call expires day 40. Let filtered $u \sim \mathcal{N}(\log 110, 0.04^2)$, $\delta = 0$, $p_{\text{on}} = 0.90, p_{\text{off}} = 0.60$, $\delta_{\text{exit}} = 0.01$. **Entry:** at $S = 100$, $p \approx 0.99$; enter only if the call wins the $\mathbb{P}$-path/$\mathbb{Q}$-mark comparison after costs. **Noise:** an intraday dip with unchanged thesis evidence does not trigger a decision or invalidate it. **Week 2:** at $S = 106$, remaining gap $\log(110/106) = 0.037$, $p \approx 0.82$: retain the thesis. If the after-cost value gain from a smaller call spread (or shares) exceeds h_0, switch; otherwise hold. **Exit:** $S = 109$ gives gap $0.009 < \delta_{\text{exit}}$, so close on convergence. Alternatively a failed catalyst moving the anchor to 102 at $S = 106$ gives $p \approx 0.17$: invalidate. Day 20 closes any remaining thesis risk; zero exposure can be optimal earlier while the thesis remains active. **Composition:** $O \to (X, y) \to (b, \widehat{\theta}, d, z) \to \mathsf{P}_k^H \to \{c^{A,Q}(Z_l^{(p)}), W_l^{(p)}\} \to (Q^H, \mathcal{A}) \to a^{\text{str}} \to \text{fills} \to W' \to b'$.

$$a_k^* = F(O_{\leq k}, s_k^{\text{acct}}; b_k, \widehat{\theta}, \widehat{\beta}_A) = \mathcal{E}_{\text{O6}}\left[\text{Gate}_{\text{O4,O8}}\left(\arg\max_{a \in \mathcal{A}(s_k)} \left\{\mathbb{E}_{\substack{Z \sim \mathsf{P}_k^H,\ C_l = c^{A,Q}(Z_l) \\ s' = \text{Ledger}_{\text{O7}}(s_k, a)}} [-\eta_T \|D_T a_{\text{trade}}\|_1 + V_{k+1}^{H-1}(s')] - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}\right\}\right)\right].$$